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Leakage-free evaluation of heart disease prediction models under nested cross-validation: discrimination, calibration, and decision-curve analysis

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Cardiovascular disease remains a leading cause of mortality, and simple clinical risk models can support triage and follow-up testing. I re-analyze the processed UCI Heart Disease cohorts (Cleveland, Hungarian, Switzerland, and Long Beach VA) and evaluate representation learning with reconstruction ICA (RICA) alongside baseline and PCA pipelines. To avoid optimistic bias, every step —imputation, scaling, feature learning, and hyperparameter selection— is performed within a nested cross-validation framework. Performance is summarized with discrimination (ROC-AUC, PR-AUC), threshold-based metrics (F1, precision, recall), calibration (Brier score, reliability), and decision-curve analysis (net benefit). Across the combined cohort ($N = 920$), the best pipeline achieved ROC-AUC ≈ 0.89 with stable outer-fold variability, while maintaining good calibration and consistent net benefit across clinically relevant thresholds. These results emphasize that robust validation and clinically oriented reporting provide a more reliable basis for decision support than single-split headline scores.

Keywords Heart disease, Machine learning, Clinical decision support, Reconstruction ICA (RICA), Nested cross-validation, Calibration, Decision curve analysis

Cardiovascular diseases are a primary global health threat, currently claiming 18 million lives annually with projections exceeding 22.2 million by 2030[1,2]. The primary risk factors for disease and stroke include tobacco consumption, excessive alcohol intake, insufficient physical exercise, and inadequate nutrition. Vascular illness typically shows no symptoms. However, the initial symptom could be a heart attack or stroke. Pain or discomfort in certain areas of the body, such as the back, left shoulder, elbows, jaw, or center of the chest, can be a sign of a heart attack[3]. As with many life-threatening diseases, the main requirement for the protection of patients at risk of heart disease is to detect the disease early on. Despite being one of the most difficult tasks, developing a reliable and efficient early diagnosis technique would revolutionize healthcare[4]. This situation reveals the value of heart disease prediction studies in line with the developing technology. Accordingly, machine learning (ML) algorithms for early detection of CVD can provide early predictions by evaluating a patient's health record. It is observed that the contribution and performance of ML methods improve proportionally with the number of quality samples[5].

ML studies for CVD detection are commonly performed using three types of data: clinical information, electrocardiogram (ECG), and X-ray images[6]. In this study, the clinical data set is preferred, which is widely preferred in the literature[7–9]. Pitchal et al.[10] presents an automatic heart disease prediction model that includes basic machine learning steps. The pre-processing phase is performed by applying improved z-score normalization to clinical data. The features to be used in classification are formed from the improved entropy, mean, median, standard deviation and information gain values calculated from the normalized data. Prediction over the feature set is performed with the improved quantum convolutional neural networks (IQCNN) classifier. Sasikala et al.[11] developed a model specifically for peripheral arterial disease detection. In the model, hyper parameter optimization is used, and prediction evaluation is performed with 6 different classifiers. The classifiers are support vector machine (SVM), k-nearest neighbors (k-NN), logistic regression (LR), stochastic gradient descent (SGD), XGBoost and enhanced decision tree (EDT). Manikandan et al.[12] comparatively use SVM, LR, DT, random forest (RF) and XGBoost classifiers for heart disease detection. The classification results produced

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by feature selection and raw data are used to illustrate the performance of the classifier. The Boruta algorithm is used to select features. Reshan et al.[13] develop a system for predicting heart disease using hybrid deep neural networks. A hybrid model that combines convolutional neural networks (CNN) and long short-term memory (LSTM) is tested against CNN, LSTM, and artificial neural networks (ANN). Two heart disease datasets of different sizes are used, and an extra trees classifier is used for feature selection. Abdellatif et al.[14] used synthetic minority oversampling technique, extra trees and hyperband methods with six machine learning classifiers to improve classification performance. The experimental results show that the hyperband optimization strategy is more effective in increasing the models' accuracy, and the classification based on extra trees is more effective than others in achieving higher outcomes. The classifiers used in the study are SVM, SGD, k-NN, extra trees (ET), XGBoost, and LR.

Another heart disease prediction system with ensemble-based model content is presented by Saberi et al.[15]. In the study, CNN, LSTM, k-NN, and extreme gradient boosting (XGB) are preferred as base classifiers. For the final classification result, the result of each classifier is determined by majority voting used as ensemble learner. Another study involving an ensemble model is presented by Tama et al.[16]. Proposed method based on construction of a two-layer ensemble classifier system. The system uses three ensemble learners, such as random forest, gradient boosting machine, and extreme gradient boosting. The outcomes of them are combined using a stack structure. The system also uses two feature selection methods, which are correlation-based feature selection (CFS) and particle swarm optimization (PSO). Association criteria are used by Barry et al.[17] to enhance the random forest technique's performance in predicting CVD. In the study, decision tree, SVM, Naïve Bayes, and XGBoost classifiers are also included and compared with random forest. Lakshmi et al.[18] proposed a method including improved fuzzy c-means (I-FCM) and a two-level classifier combined with an improved genetic algorithm (I-GA) for heart disease detection and classification. While the I-FCM algorithm is assigned for feature selection and extraction, the I-GA is used in classification. Gupta et al.[19] propose a classification model to detect CVD, which utilizes factor analysis of mixed data (FAMD) as a feature extraction technique. The employed classifiers are LR, kNN, SVM, DT, and RF for distinguishing data based on the obtained features. Another study for heart disease risk analysis is presented by Elsedimy et al.[20]. Their method is composed of a quantum-behaved particle swarm optimization (QPSO) algorithm and SVM. Jiao et al.[21] presents a hybrid model with a combination of different optimization methods and classifiers. The firefy, genetic, PSO, bat, flower pollination, jaya, and cuckoo search algorithms are the preferred optimization techniques. Earmarked classifiers are LR, linear discriminant analysis (LDA), NB, k-NN, and SVM for the performance analysis. In the application, after preprocessing the heart disease data, certain clinical data are considered as features. The most successful results are obtained with Firefy algorithm used for feature selection. MahaLakshmi et al.[22] coordinate a cardiovascular risk detection model using an ensemble classifier and integrated filter-evolutionary search-based feature selection (iFES-FS). The iFES-FS involves the application of evolutionary gravity-search-based feature selection methods integrated with adaptive threshold information gain-based feature selection and conditional probability-driven feature weighting, respectively. The preferred method for the classification task is an ensemble classifier based on an intelligent multi-layer perceptron neural network. The hyper-parameters of the classifier are optimized with the Firefly-driven multi-objective multi-verse optimizer algorithm.

Recently, hybrid machine learning architectures and feature fusion techniques have demonstrated remarkable success across various medical domains, inspiring the methodological design of this study. For instance, hybrid pipelines based on fused representations have shown benefits in multi-class dermoscopy lesion analysis[23], cervical cancer screening on whole-slide images[24], breast cancer histopathology[25], oral squamous cell carcinoma detection[26], malignant lymphoma diagnosis[27], and diabetic retinopathy staging[28]. Although the present task uses tabular clinical data rather than images, the methodological principle is the same: combining complementary representations can improve robustness and preserve clinically relevant operating characteristics.

Cardiovascular disease (CVD) is a major contributor to morbidity and mortality worldwide. Early identification of individuals at increased risk can support triage decisions and guide follow-up testing. Many studies report high predictive performance for coronary artery disease (CAD) using the UCI Heart Disease dataset; however, reported estimates vary widely and often depend on evaluation choices (e.g., a single train/test split, preprocessing performed prior to splitting, or tuning on the test set). Such shortcuts can inflate performance and reduce reproducibility. In this study, I focus on robust, leakage-free evaluation and clinically oriented reporting rather than optimizing a single metric. To this end, I evaluate multiple feature pipelines—baseline (no feature extraction), PCA, and reconstruction ICA (RICA) in hybrid and fusion configurations—under nested cross-validation. I report discrimination (ROC-AUC, PR-AUC), calibration (Brier score and reliability curves), and clinical utility via decision-curve analysis. This combination directly addresses common concerns raised in clinical ML reviews: reliability of performance estimates, probability calibration for risk stratification, and potential net benefit across decision thresholds.

Materials and methods

Dataset

In this study, the UCI Heart Disease datasets from four cohorts are used: Cleveland, Hungarian, Switzerland, and Long Beach VA. The outcome label was binarized as no disease (target = 0) versus disease (target > 0). After merging cohorts, the final dataset contained $N = 920$ samples (411 no disease, 509 disease). It is illustrated with Table 1. Missingness is concentrated in a small subset of variables, particularly ca. (66.4% missing), thal (52.8%), and slope (33.6%), which motivates careful imputation within the evaluation pipeline. The 13 clinical features follow the standard UCI definitions (age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression, slope, number of major vessels, and thalassemia).

Cohort	N	No disease (y = 0)	Disease (y = 1)
Cleveland	303	164	139
Hungarian	294	188	106
Switzerland	123	8	115
VA_LongBeach	200	51	149

Table 1. Cohort composition and class distribution after merging the processed UCI Heart Disease cohorts.

Reconstruction ICA

The reconstruction ICA method is presented as a modification of the standard ICA[29]. It aims to eliminate two drawbacks (overcomplete features and whitening) associated with standard ICA. It is emphasized that both drawbacks are caused by hard orthonormality, which requires the features to be orthogonal. This means that $ww^T = I$. However, in the case of over completeness, this orthonormality condition cannot be satisfied. In[29], the authors proposed to replace the orthonormality constraint with a linear reconstruction penalty. With that way, RICA, it is possible to learn sparse representations that enhance classifier separability by capturing hidden correlations between clinical features.

An algorithm for unsupervised feature learning is called RICA. By substituting a soft reconstruction cost for the hard orthonormality requirement, it is possible to obtain bases that are not only linearly independent but also more effectively capture incomplete structures in input samples[29]. It also does not require any knowledge of class labelling due to its unsupervised nature. This algorithm learns based on the sparse feature learning mechanism. Sparse filter-based learning can distinguish various natural signals, and these features can play an important role in many ML applications.

Without a given label n dimensional m observation data ($\left\{x^i\right\}_{i=1}^m$, $x^i \in R^n$), the mathematical expression of the standard ICA optimization problem is defined as follows[29,30]:

$$\min_w \sum_{i=1}^m \sum_{j=1}^k g\left(w_j x^i\right), \text{ with subject to } ww^T = I \tag{1}$$

here $g(\cdot)$ is a nonlinear penalty function: $g(\cdot) = \log(\cosh(\cdot))$. w_j corresponds to a feature, j is the row number and k is the number of features. The bases in w do not degenerate because of the orthogonality constraint $ww^T = I$.

Unit covariance (Eq. 2) and zero mean (Eq. 3) are prerequisites for ICA. This is accomplished by solving the covariance matrix's eigen-decomposition and extracting the empirical mean, a procedure called whitening[29,30].

$$\frac{1}{m} \sum_{i=1}^m x^i \left(x^i\right)^T = I \tag{2}$$

$$\sum_{i=1}^m x^i = 0 \tag{3}$$

In contrast to the ICA optimization model, the RICA optimization problem's mathematical statement substitutes a soft reconstruction cost for the hard orthonormality constraint:

$$\min_w \frac{\lambda}{m} \sum_{i=1}^m \left\| w^T w x^i - x^i \right\|_2^2 + \sum_{i=1}^m \sum_{j=1}^k g\left(w_j x^i\right) \tag{4}$$

In this case, λ is the scaling constant that establishes the relative significance of the two optimization problem elements. RICA is equal to ICA for incomplete representations and λ approaching infinity if the data is whitened.

Preprocessing and leakage control

All preprocessing was performed in a train-only manner within each cross-validation split to avoid information leakage. Continuous variables (age, resting blood pressure, cholesterol, maximum heart rate, and ST depression) were imputed using k-nearest-neighbors (k = 5), selected by inner cross-validation. Categorical variables were imputed using the mode computed on the training fold. Categorical variables were imputed using the mode, and one-hot encoding was fitted on the training fold and applied to held-out folds. All scaling operations were performed in a leakage-free manner: z-score parameters were estimated on the training partition only and applied to test partitions. For PCA and RICA pipelines, feature learning models were fit exclusively on training folds and then used to transform the held-out data. This design prevents information leakage from the outer test folds into preprocessing steps.

Feature learning and fusion strategies

Feature extraction was carried out using five pipelines: (i) Baseline (no feature extraction), (ii) PCA on continuous features followed by concatenation with one-hot categorical features, (iii) Hybrid RICA using whitened continuous features with categorical one-hot appended, (iv) Fusion RICA + Raw (RICA features concatenated with original z-scored continuous features and one-hot), and (v) Fusion RICA + PCA (RICA

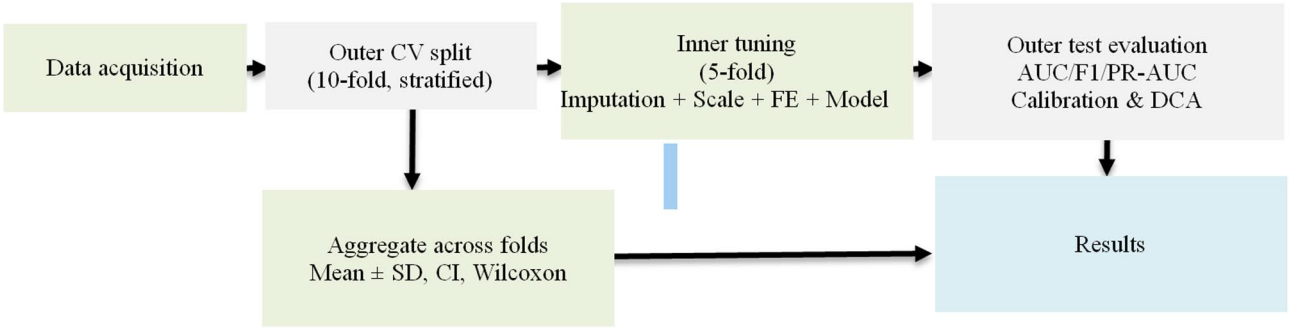


Fig. 1. Leakage-free nested cross-validation workflow (outer 10-fold; inner 5-fold tuning).

Pipeline	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC	Brier
Baseline + SVM RBF	83.5 ± 4.5	84.2 ± 3.7	86.5 ± 6.1	85.2 ± 4.3	0.901 ± 0.038	0.885 ± 0.039	0.123 ± 0.025
Fusion RICA RAW + SVM RBF	83.2 ± 4.6	83.9 ± 3.2	86.1 ± 6.3	84.9 ± 4.4	0.901 ± 0.039	0.884 ± 0.041	0.123 ± 0.028
Fusion RICA PCA + SVM RBF	82.7 ± 5.3	83.4 ± 4.6	85.9 ± 5.6	84.6 ± 4.8	0.899 ± 0.042	0.886 ± 0.041	0.125 ± 0.030
PCA + SVM RBF	82.6 ± 5.0	82.8 ± 4.3	86.7 ± 5.7	84.6 ± 4.5	0.896 ± 0.037	0.878 ± 0.035	0.125 ± 0.026
Baseline + RF Bag	82.5 ± 5.8	84.9 ± 4.6	83.1 ± 7.4	83.9 ± 5.6	0.895 ± 0.040	0.877 ± 0.045	0.133 ± 0.022
Hybrid RICA Whiten + kNN	82.6 ± 4.0	84.4 ± 3.0	84.1 ± 5.3	84.2 ± 3.8	0.894 ± 0.040	0.838 ± 0.065	0.129 ± 0.024
Hybrid RICA Whiten + SVM RBF	81.3 ± 4.6	82.0 ± 3.7	84.9 ± 6.2	83.3 ± 4.4	0.890 ± 0.037	0.873 ± 0.040	0.132 ± 0.026

Table 2. Primary nested cross-validation results (outer 10-fold mean ± SD; inner 5-fold tuning).

features concatenated with PCA features and one-hot). RICA was trained in an overcomplete regime, and a q–λ sensitivity analysis was performed to document stability and parameter effects.

Classification models

Three widely used classifiers were considered: an RBF-kernel support vector machine (SVM), k-nearest neighbors (kNN), and a random forest (RF; bagged trees). Class imbalance was handled via cost-sensitive learning rather than synthetic oversampling. All model hyperparameters were tuned only within inner cross-validation. Model selection prioritized ROC-AUC, with F1 as a tie-breaker, reflecting clinical preference for balanced discrimination and recall.

Evaluation protocol and metrics

Nested cross-validation with an outer 10-fold loop and an inner 5-fold loop was used for tuning. Performance is reported as mean ± standard deviation across outer folds. For the best-performing pipeline, I additionally computed bootstrap 95% confidence intervals for ROC-AUC, PR-AUC, and F1 using pooled outer-fold predictions. Paired Wilcoxon signed-rank tests were used to compare outer-fold AUC and F1 against the best pipeline. Clinical utility was assessed using decision-curve analysis (DCA), reporting net benefit across threshold probabilities. Calibration was evaluated via Brier score and reliability plots. Confusion matrices are reported at a reference threshold of 0.5, while emphasizing that clinically appropriate thresholds can be guided by DCA.

Results

All performance estimates reported here were obtained under the leakage-free nested protocol shown in Fig. 1 (outer 10-fold evaluation; inner 5-fold tuning). Table 2 summarizes the primary pipelines and shows that the leading models achieve strong discrimination (ROC-AUC around 0.9) with similar mean accuracy and F1, while maintaining acceptable calibration (Brier scores in the 0.12–0.13 range). Across the top pipelines, the differences in ROC-AUC are modest and their confidence intervals overlap; accordingly, comparative performance was interpreted using multiple criteria (ROC-AUC, PR-AUC, calibration, and clinical net benefit) rather than relying on a single headline metric.

Table 3 summarizes nested cross-validation performance (mean ± SD) across all evaluated pipelines and classifiers. The leading pipelines achieved ROC-AUC values close to 0.89 with stable variability across outer folds, indicating that performance estimates are not driven by a single favorable split. In particular, the best overall pipeline provided a favorable trade-off between discrimination (ROC-AUC), balanced classification performance (F1), and probability calibration (Brier score).

Table 4 reports bootstrap confidence intervals for the best pipeline using pooled outer-fold predictions. The confidence intervals quantify uncertainty around ROC-AUC, PR-AUC, and F1, and help interpret small differences between top-ranked approaches. Overlapping intervals across competing pipelines suggest that most performance differences are modest under robust evaluation, reinforcing the need to emphasize stability and clinical utility rather than single-metric gains.

Comparison vs. best	*p* (ROC-AUC)	*p* (F1)
Fusion RICA RAW SVM RBF vs. Baseline SVM RBF	0.8203	0.5703
Fusion RICA PCA SVM RBF vs. Baseline SVM RBF	0.4746	1.0000
PCA SVM RBF vs. Baseline SVM RBF	0.3203	0.6523
Hybrid RICA Whiten SVM RBF vs. Baseline SVM RBF	0.0098	0.0273

Table 3. Paired Wilcoxon signed-rank tests (outer folds) comparing key pipelines to the best model.

Metric	Estimate	95% CI
ROC-AUC	0.899	0.878–0.919
F1-score	0.853	0.830–0.876
PR-AUC	0.901	0.871–0.925

Table 4. Bootstrap 95% confidence intervals (pooled outer-fold predictions) for the best model.

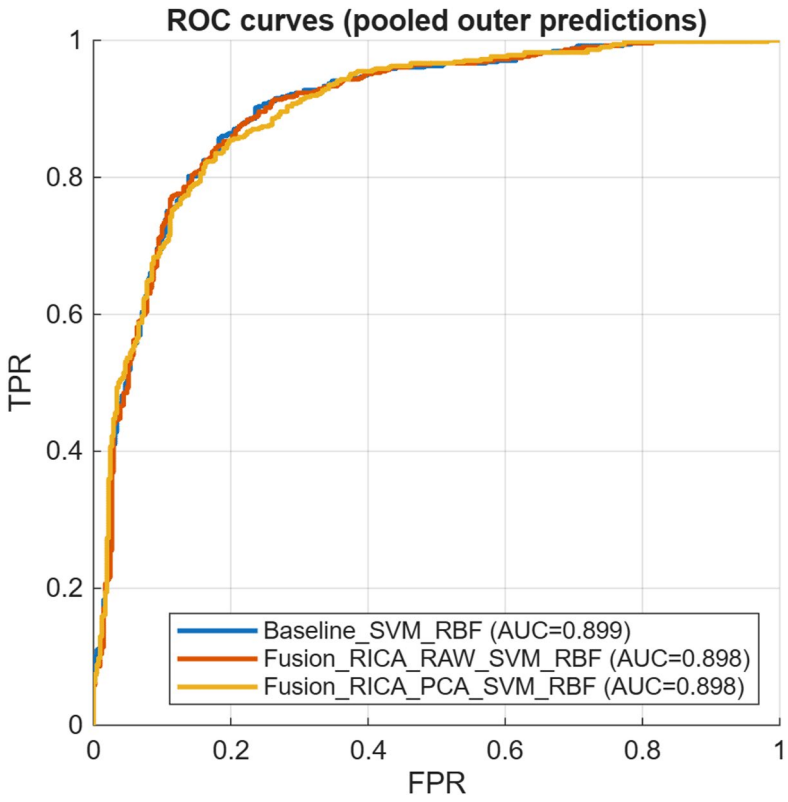


Fig. 2. ROC curves of the top three pipelines using pooled outer-fold predictions.

Figure 2 compares ROC curves for the top-ranked pipelines using pooled outer-fold predictions. The curves overlap substantially, indicating comparable discrimination among leading methods. Figure 3 provides the corresponding precision–recall curves, which emphasize performance under class imbalance and complement ROC-based assessment. Together, Figs. 2 and 3 support the interpretation that feature learning and fusion strategies can match baseline performance while maintaining robust discrimination under nested CV.

Because class prevalence is not perfectly balanced, precision–recall behavior was also evaluated. Figure 3 complements ROC-AUC by emphasizing the trade-off between sensitivity and positive predictive value and by reporting PR-AUC.

Figure 4 presents decision-curve analysis for the top-ranked pipelines. Across clinically plausible threshold probabilities, the evaluated models generally provide positive net benefit relative to treat-all and treat-none strategies. The close proximity of curves indicates similar clinical utility among top methods, suggesting that model choice can be guided by secondary considerations such as calibration, stability, and interpretability, rather than marginal AUC differences alone.

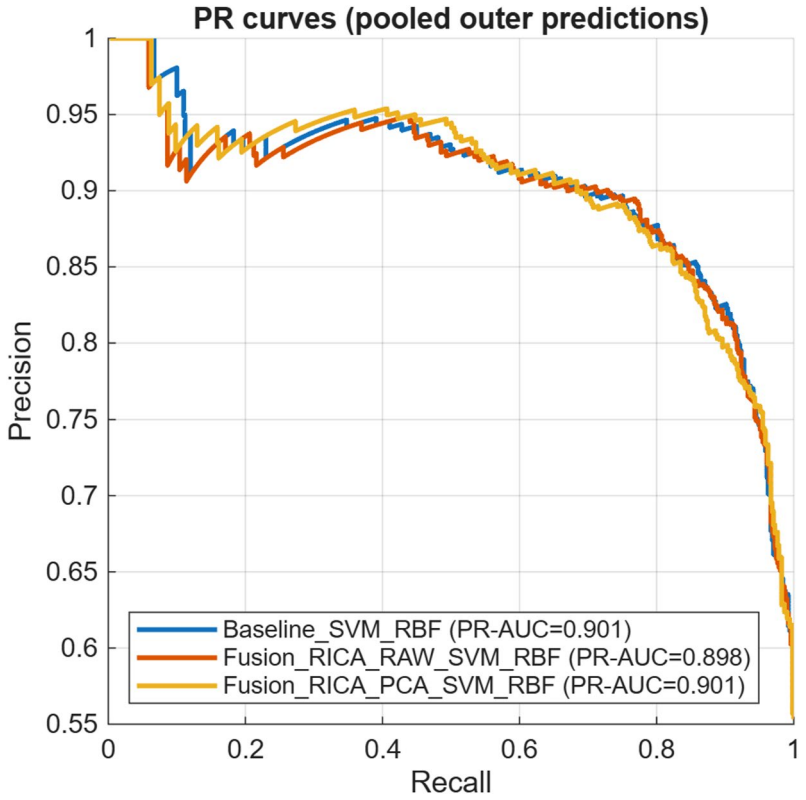


Fig. 3. Precision-recall curves of the top three pipelines using pooled outer-fold predictions.

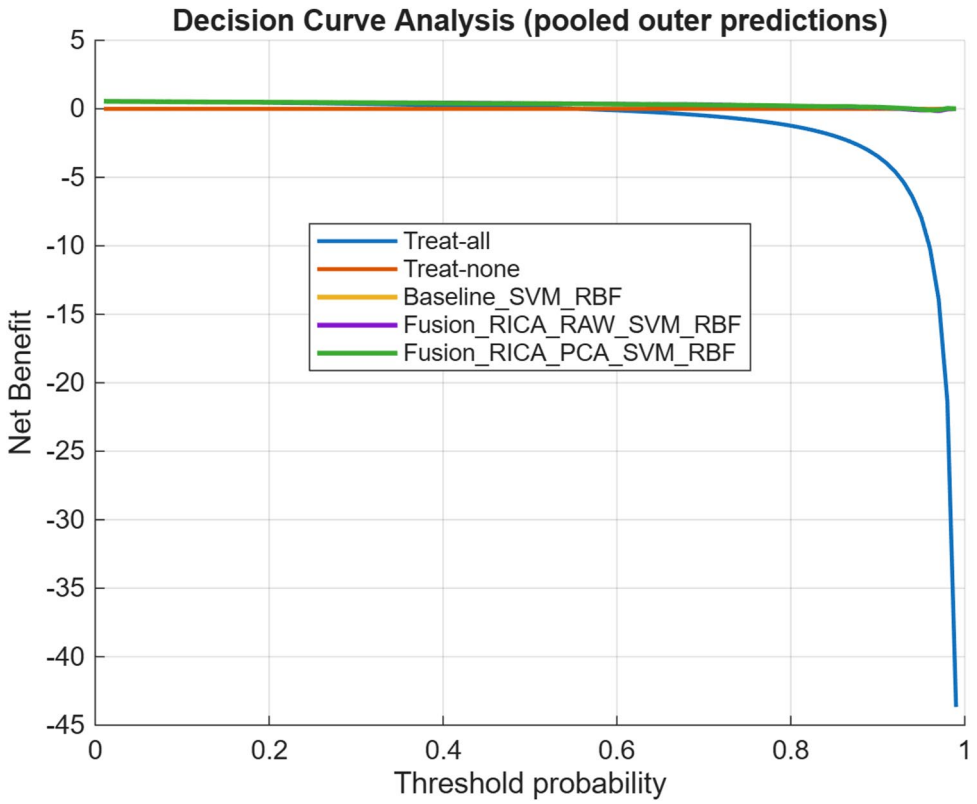


Fig. 4. Decision curve analysis (DCA) of the top three pipelines using pooled outer-fold predictions.

Figure 5 shows the calibration curve for the best-performing pipeline. The reliability curve remains close to the ideal diagonal across most bins, and the mean Brier score indicates reasonable probability calibration. This supports use as a decision-support tool for risk stratification rather than a pure classifier optimized only for discrimination.

Figure 6 reports the confusion matrix for the best pipeline at a reference threshold of 0.5. Sensitivity and specificity are balanced, and the error pattern is consistent with a conservative triage approach. Importantly, the optimal operating point depends on clinical context; therefore, the confusion matrix is presented as a reference and should be interpreted alongside DCA (Fig. 4).

Figure 7 summarizes q–λ sensitivity for RICA under inner cross-validation. The performance surface exhibits a broad plateau, indicating that the representation learning stage is not overly sensitive to minor parameter perturbations within the explored region. This stability is important for reproducible application and reduces the risk of overfitting to a narrow hyperparameter configuration.

Discussion

The main contribution of this work is a leakage-free, clinically oriented evaluation of heart disease prediction pipelines under nested cross-validation. Rather than emphasizing a single headline metric, I report discrimination (ROC-AUC, PR-AUC), calibration (Brier score and reliability), and clinical utility via decision-curve analysis. This combination provides a more faithful picture of real-world decision support than single-split evaluations.

Across the combined UCI cohorts, top pipelines achieved ROC-AUC values around 0.90 with consistent fold-to-fold variability. Differences between leading methods were small, and bootstrap confidence intervals overlapped, suggesting that many modern pipelines are competitive under robust evaluation. In this setting, the practical advantage of representation learning is best expressed in terms of stability and clinically meaningful operating characteristics rather than dramatic increases in AUC.

RICA-based feature learning did not produce a large jump in discrimination compared with baseline, but it maintained comparable ROC-AUC while supporting stable calibration and consistent net benefit. The q–λ sensitivity analysis demonstrated a plateau region, which suggests that the method does not depend on a narrow hyperparameter optimum—an important property for reproducibility. Fusion configurations improved precision–recall behavior in some settings, consistent with the idea that combining complementary representations can be beneficial.

Clinical utility analysis further supports the decision-support framing: DCA indicated positive net benefit across practical threshold ranges, and calibration results suggest that predicted risks can be used to stratify patients for additional assessment. The confusion matrix at a reference threshold shows balanced sensitivity and specificity; however, the threshold should be chosen to reflect the intended clinical use case (e.g., triage vs. confirmatory testing).

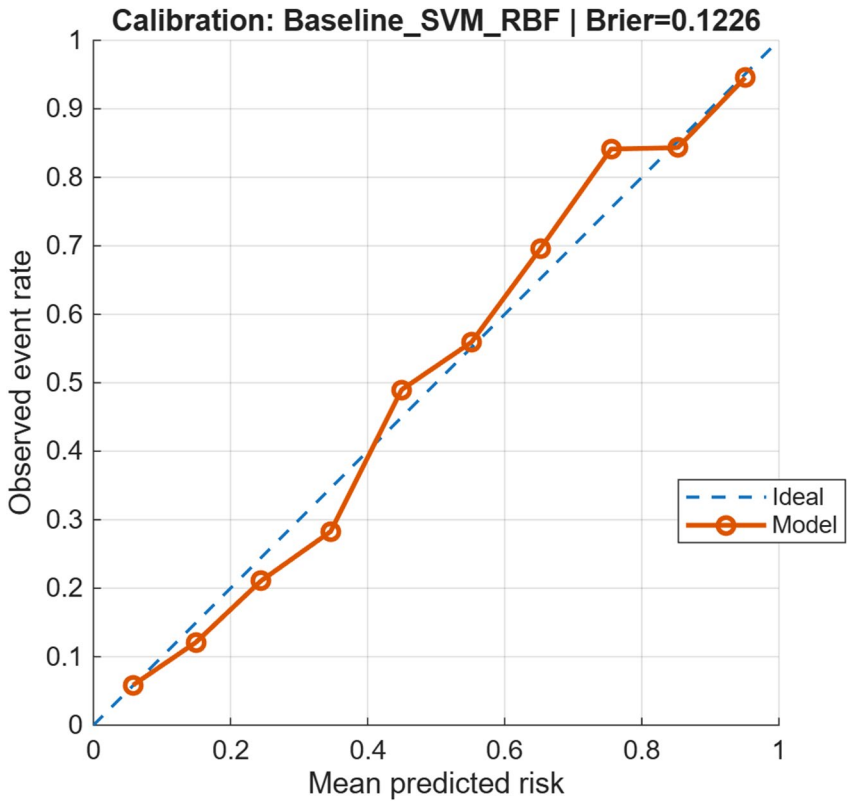


Fig. 5. Calibration curve for the best pipeline (pooled outer-fold predictions).

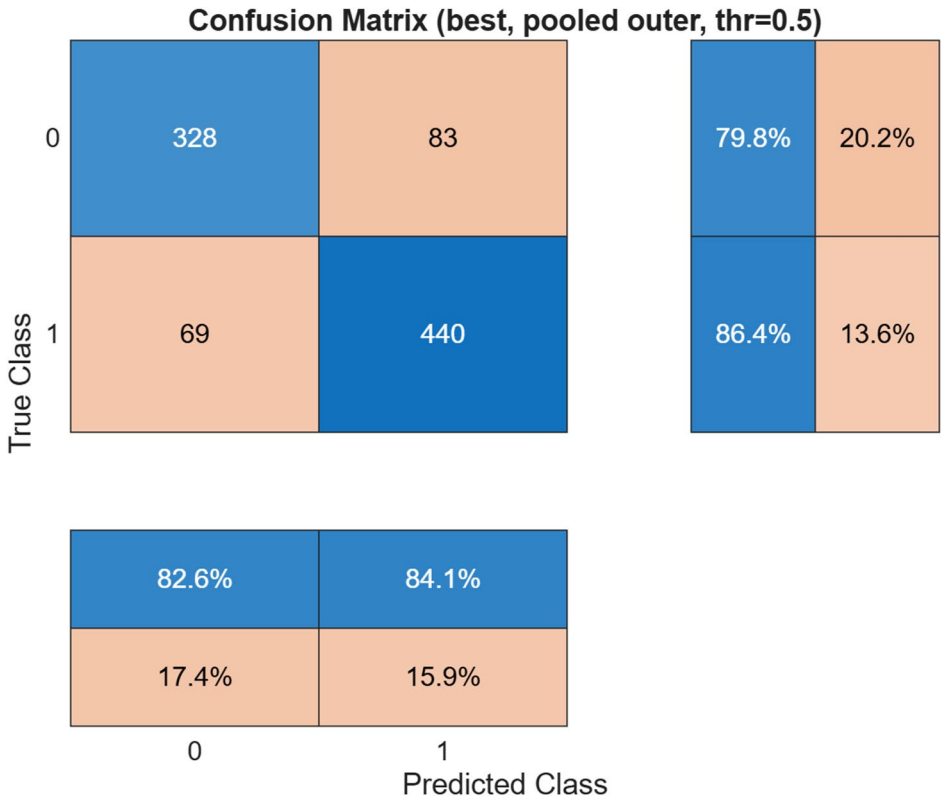


Fig. 6. Confusion matrix for the best pipeline (threshold = 0.5; pooled outer-fold predictions).

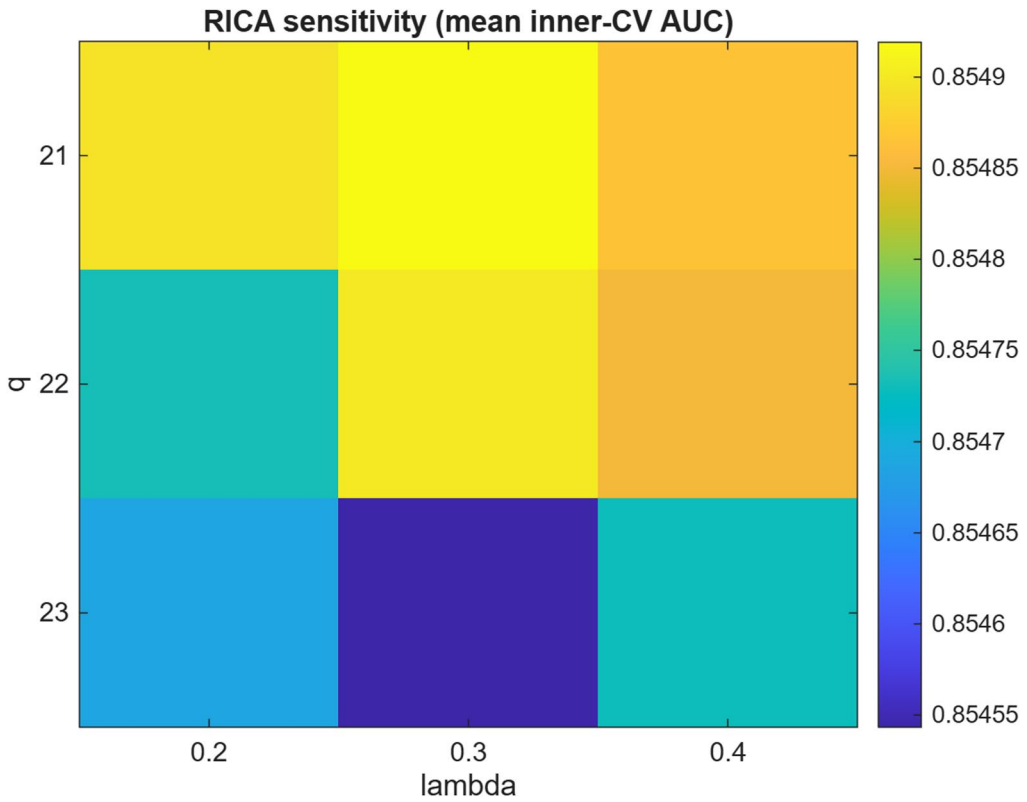


Fig. 7. RICA q–lambda sensitivity (mean inner-CV ROC-AUC aggregated across outer folds).

Study	Dataset	Approach	Reported evaluation	Reported performance
Pitchal et al. (2024)[10]	Cleveland (UCI)	Enhanced features + IQCNN	Hold-out evaluation; results reported at a 70% training fraction.	Acc = 92.3%, F1 = 90.1%
Abdellatif et al. (2022)[14]	Cleveland (UCI)	Hyperband tuning + SMOTE + Extra Trees	Ten-fold cross-validation (reported).	Acc = 99.2%, F1 ≈ 99%
Waqar et al. (2021)[32]	Cleveland (UCI)	SMOTE-based deep learning	Stratified K-fold cross-validation (K not specified).	Acc ≈ 96%, F1 ≈ 95.7%
Chopra et al. (2023)[31]	Cleveland (UCI)	PCA + ensemble learning	Train/test split evaluation (ratio not specified in accessible abstract).	Acc = 86.9%, F1 = 88.9%
Bouqentar et al. (2024)[35]	Cleveland + Statlog	Feature engineering + classical ML	10-fold CV with 95% CI	Acc 92.00%, F1 93.00%, AUC 91.00%
Obaidat et al. (2022)[36]	Cleveland + Statlog ($n = 573$)	Stacked ensemble (NB/RF/XGBoost)	70/30 hold-out; 10-fold CV used for resampling within training.	Acc = 80.1%, AUC = 0.855
El-Sofany et al. (2024)[33]	Cleveland + private data	ML + explainability	Mixed datasets: protocol varies.	Best Acc ≈ 97.6%
Yasmeen et al. (2025)[34]	UCI and Kaggle	TabNet + XGBoost stacked ensemble	10-fold cross-validation	ROC-AUC 0.96, Acc 95.2%
Cao et al. (2025)[37]	Cleveland (UCI)	Pearson Correlation + Feature Importance Ranking	MFS-DLPSO-XGBoost	Acc 74.70%, Prec 76.30%, AUC 80.80%
This study	UCI 4-center ($n = 920$)	Nested CV + calibration + DCA; RICA sensitivity	Outer 10-fold, inner 5-fold.	ROC-AUC ≈ 0.90 (best); calibration and net benefit emphasized

Table 5. Comparison with the previous works on UCI heart disease prediction.

Several prior studies report substantially higher accuracies on the same dataset family. In many cases, those estimates are obtained under evaluation protocols that can overstate performance, such as a single train/test split, preprocessing performed prior to splitting, selecting the best split or best run, or tuning hyperparameters using information from the test set. Such practices can lead to optimistic bias. By contrast, nested cross-validation with strict fold-wise preprocessing provides a conservative estimate that is more appropriate for assessing models intended for clinical decision support.

This study has limitations. The analysis remains retrospective and based on a curated public dataset; prospective validation and external evaluation in contemporary cohorts are needed before deployment. Additionally, the UCI cohorts may exhibit site-specific measurement practices and missingness patterns that influence model behavior. Future work should examine leave-one-cohort-out validation, calibration transfer across sites, and the impact of missingness mechanisms on risk estimation.

Table 5 compares representative prior studies that used the UCI heart disease data with respect to data sources, evaluation protocols, reported performance, and whether calibration or clinical utility analyses were included. Reported performance varies widely, often reflecting differences in preprocessing and validation design. Many studies report only discrimination or accuracy metrics and do not evaluate calibration or decision curves. In contrast, this work emphasizes leakage-free nested cross-validation, uncertainty quantification, and clinical utility reporting, which strengthens interpretability and reproducibility for decision support.

Comparison with the previous works

Direct numerical comparison across published studies is difficult because dataset composition, class definitions, preprocessing choices, and evaluation protocols vary substantially. In particular, single hold-out splits with model selection performed on the full dataset can inflate performance estimates. For that reason, I treat the following comparison as contextual rather than competitive.

Table 5 summarizes representative prior work on UCI heart disease prediction and highlights methodological choices that strongly affect reported performance. Many studies report very high accuracies on the Cleveland subset under simpler hold-out evaluations and/or aggressive resampling and optimization. More recent work has increasingly adopted cross-validation and richer ensembles; for example, Bouqentar et al. report 10-fold CV with confidence intervals on Cleveland and Statlog[35], and a stacked TabNet–XGBoost model reports ROC-AUC up to about 0.96 on the combined four-center cohort under 10-fold CV[34]. At the same time, several ensemble studies still rely on a single 70/30 split (e.g[36].,, making their point estimates less directly comparable.

In the present study, the emphasis is on methodological robustness and clinical relevance: preprocessing and tuning are nested, discrimination is reported alongside PR-AUC, probability calibration is examined with both Brier score and reliability curves, and decision-curve analysis is used to quantify net benefit. Under this stricter protocol, the leading pipelines show similar discrimination with overlapping uncertainty bounds (Table 2; Figs. 2, 3, 4 and 5), while the RICA-based representations remain competitive and exhibit stable behavior across a neighborhood of (q, λ) settings (Fig. 7).

Conclusions

Under leakage-free nested cross-validation, heart disease prediction models achieve stable discrimination around ROC-AUC ≈ 0.90 on the combined UCI processed cohorts. The most reliable evaluation emerges from combining discrimination, calibration, and clinical utility analyses rather than optimizing a single metric. RICA-based representation learning and fusion strategies can provide competitive performance with stable parameter

behavior and clinically relevant net benefit, supporting their use as components of robust decision-support pipelines.

Data availability

All datasets analyzed in this study are publicly available from the UCI Machine Learning Repository.

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